

```
kali@kali: ~  
Session Actions Edit View Help  
Preparing to unpack .../6-gnupg_2.4.8-4_all.deb ...  
Unpacking gnupg (2.4.8-4) over (2.4.8-3) ...  
Preparing to unpack .../7-gpgsm_2.4.8-4+b1_amd64.deb ...  
Unpacking gpgsm (2.4.8-4+b1) over (2.4.8-3) ...  
Preparing to unpack .../8-gnupg-l10n_2.4.8-4_all.deb ...  
Unpacking gnupg-l10n (2.4.8-4) over (2.4.8-3) ...  
Preparing to unpack .../9-gpgv_2.4.8-4+b1_amd64.deb ...  
Unpacking gpgv (2.4.8-4+b1) over (2.4.8-3) ...  
Setting up gnupg-l10n (2.4.8-4) ...  
Setting up gpgv (2.4.8-4+b1) ...  
Setting up gpgconf (2.4.8-4+b1) ...  
Setting up gpg (2.4.8-4+b1) ...  
Setting up gnupg-utils (2.4.8-4+b1) ...  
Setting up gpg-agent (2.4.8-4+b1) ...  
Setting up gpgsm (2.4.8-4+b1) ...  
Setting up dirmngr (2.4.8-4+b1) ...  
Setting up gnupg (2.4.8-4) ...  
Setting up gpg-wks-client (2.4.8-4+b1) ...  
Processing triggers for kali-menu (2025.3.2) ...  
Processing triggers for man-db (2.13.1-1) ...  
Scanning processes...  
Scanning candidates...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
User sessions running outdated binaries:  
kali @ user service: gpg-agent.service[1346]  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
  
(kali@kali)-[~]  
$ gpg --version  
gpg (GnuPG) 2.4.8  
libgcrypt 1.11.2  
Copyright (C) 2025 g10 Code GmbH  
License GNU GPL-3.0-or-later <https://gnu.org/licenses/gpl.html>  
This is free software: you are free to change and redistribute it.  
There is NO WARRANTY, to the extent permitted by law.  
  
Home: /home/kali/.gnupg  
Supported algorithms:  
Pubkey: RSA, ELG, DSA, ECDH, ECDSA, EDDSA  
Cipher: IDEA, 3DES, CAST5, BLOWFISH, AES, AES192, AES256, TWOFISH,  
CAMELLIA128, CAMELLIA192, CAMELLIA256  
Hash: SHA1, RIPEMD160, SHA256, SHA384, SHA512, SHA224  
Compression: Uncompressed, ZIP, ZLIB, BZIP2  
  
(kali@kali)-[~]  
$
```

```

gpg: keybox '/home/kali/.gnupg/pubring.kbx' created
Please select what kind of key you want:
  (1) RSA and RSA
  (2) DSA and Elgamal
  (3) DSA (sign only)
  (4) RSA (sign only)
  (9) ECC (sign and encrypt) *default*
  (10) ECC (sign only)
  (14) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid.
  0 = key does not expire
  <n> = key expires in n days
  <n>w = key expires in n weeks
  <n>m = key expires in n months
  <n>y = key expires in n years
Key is valid for? (0) 1y
Key expires at Thu 17 Dec 2026 10:30:41 AM EST
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: ZanS Test
Email address: zans@kv.test
Comment: KV-Vaje
You selected this USER-ID:
  "ZanS Test (KV-Vaje) <zans@kv.test>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: /home/kali/.gnupg/trustdb.gpg: trustdb created
gpg: directory '/home/kali/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/kali/.gnupg/openpgp-revocs.d/9EEDFB1542D4D8D0916EB1A18DF495E7B56EC5CD.rev'
public and secret key created and signed.

pub   rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
       9EEDFB1542D4D8D0916EB1A18DF495E7B56EC5CD
uid           [ultimate] ZanS Test (KV-Vaje) <zans@kv.test>
sub   rsa4096 2025-12-17 [E] [expires: 2026-12-17]

(kali@kali)-[~]
$

```

```

(kali@kali)-[~]
$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3  completes needed: 1  trust model: pgp
gpg: depth: 0  valid: 1  signed: 0  trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2026-12-17
/home/kali/.gnupg/pubring.kbx
-----
pub   rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
       9EEDFB1542D4D8D0916EB1A18DF495E7B56EC5CD
uid           [ultimate] ZanS Test (KV-Vaje) <zans@kv.test>
sub   rsa4096 2025-12-17 [E] [expires: 2026-12-17]

```

```
(kali㉿kali)-[~/vaje14]
```

```
$ gpg --list-keys
```

```
gpg: checking the trustdb
```

```
gpg: marginals needed: 3  completes needed: 1  trust model: pgp
```

```
gpg: depth: 0  valid: 2  signed: 0  trust: 0-, 0q, 0n, 0m, 0f, 2u
```

```
gpg: next trustdb check due at 2026-12-17
```

```
/home/kali/.gnupg/pubring.kbx
```

```
pub  rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
```

```
9EEDFB1542D4D8D0916EB1A18DF495E7B56EC5CD
```

```
uid          [ultimate] Zans Test (KV-Vaje) <zans@kv.test>
```

```
sub  rsa4096 2025-12-17 [E] [expires: 2026-12-17]
```

```
pub  rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
```

```
2C88FDFF21C89B21CD491009E9AA2BDF0968E3AB
```

```
uid          [ultimate] Zans2 Test (Vaje2) <zankv2@kv.test>
```

```
sub  rsa4096 2025-12-17 [E] [expires: 2026-12-17]
```

```
(kali㉿kali)-[~/vaje14]
```

```
$
```

```

(kali㉿kali)-[~/vaje14]
$ echo "To: peer@example.com
From: student@example.com
Date: $(date)
Secret message: Zaupno sporocilo" > message.txt

(kali㉿kali)-[~/vaje14]
$ cat message.txt
To: peer@example.com
From: student@example.com
Date: Wed Dec 17 10:48:54 AM EST 2025
Secret message: Zaupno sporocilo

(kali㉿kali)-[~/vaje14]
$ gpg --encrypt --sign --armor --recipient zans@kv.test message.txt

(kali㉿kali)-[~/vaje14]
$ cat message.txt.asx
cat: message.txt.asx: No such file or directory

(kali㉿kali)-[~/vaje14]
$ cat message.txt.asc
-----BEGIN PGP MESSAGE-----

hQIMA+jcSi42YYNOAQ/+PT854ZC53WXIMuLUsvUj7MVT7Pj1HkqaGGeQ0Jy1SfoX
2+4X0F/YBYZfT1J7iHPR0mnIEfynqhvWwCGk79aFlzTORrb+wYUUGxU2Iyf5eibv
Sk3LMQpVm7Fq6Wgy0kat2QNz/seCyNZZRip05DnIJJUiFVySo0LNzQc8f2suEkQQ
OXqbVBhnUUC3ULIaZIR1C02AP2HmuSx6PRYH5k0a1kg9zpIdKNUZjSSmdqXtQMct
FW5UDR1EwXvvdOCRHN/Jw6fgo6LnyHUC9DolV68ch7NV165TStH5QdIy3TgU34if
lcw+06V/ZQoPUPc/TzMZesCG6nUfl1DYf4R1UH2Bz1FA1a4PCSLy4MmDwtDtaAwK
UBL0RUB6hVHjq34Hwur+1if0sZgVgZrvUPn4PAwf25MkMqduSp94H8Dn1/XqlhgX
X6vckPZR+VEsKK5f2/uYLw+kPk190zIXz6gwfL/McEU8wqkZ2WSn9Gemqrrd4qUi
qP+z4DZjY2984zN2gSBmqeOGtZiQPAootETVwoSBOTL/ScgGWdtL/TvdX6ZQo9Qz
TOK8S4LhCvdAH9GLsVZfp4DmXqz9LGLjI5wTaeRH5/gS66bWMG5BbtVx3V/S6JXh
l7NU39qBq4q3kU6/o+mj5c8SbNpcbzunHCmu3hjsOqSunhNj8uDcADPDpRpupKPS
6QFLCL5bfp3E7ZR2Q9QP0gWQp8k+0mydwg0PI6TyrjRHCHj68/7vRVQGc0wg2Vl8
2kqiYHVfpUvqnB2aSmLJaNEbw830K6+E842kvvrK4X9VbaS360rIUec+TPwQ/URG
DYyJUAouEbP76z/a813MzNYqcfbYnLPwsItTyLD2bGMAhOmuXSMG2xfF0/h8gXPY
0j55na58VF2DHmpmk+5LXRogBWE8YdvCxZc0BGhRSMix97JrbPowqfff9qVGDw
ygZysxl8N1CxLGmTTetIvFRIwaB4LCaafDm6E2jxikuaJK0+HFE/2A4gpsC0K0YB
tR6xPKDPD4NFNzXfB5VV9FR50m9EOQ1oRbQIZ127g6FuDuJkCs4ybInm7HPQy9rs
Qu33l0sRBe02xH6FTzXSUAGMJzlJiMaQ0amt/OAJ/sR0IYSP+9s21fU+y6l+yh3i
so15RwGc+lHljxjGE3LuDIg7agc4XP6npYgHrck5lvihAFh04RURHedc8bcFzp
WP9EcPyx//WJX3W7+Ipr2Xr1TzznRNxyPBStwNJYXDVMcNqksA+00QZIRNIAJbee
QMPT7YHhxZB0XJru8WvMGTMteOnXw7TplvenGVq0WdrUC5cTuADZPjza6yEg7Aca
8B4ds6ywdE1IHVNKgr12IOSHfKu15BUf2LI80fAipbJwEIqR/sx8rhJQrwg/Lz8
IEnTfRdie4Zbvpss2yUcU1T10CBpGvGLhVSLYZnvn22WTKSnX3HMemVo4rWfq9Zk
Exwm0S03RvKHK8zszqwmTkOvhzZ6nnSQH+mCK+fcdH4BfwqrEEIyK9W4eDB03Hoc
HH1HwGU3Z/zKupLJne+pKzG/PSw2+E/zshFHVYclzNJUVfbyQYB2sWeXEex0/Fxo
3ExSNn7daIREOCxKmk0+q9DCS9QxMeAOZY4IrPwcb7cMwAU00Z5i9Py1hMEQd7FX
iyG0wREngH7ylN+bD4tH8FyFANXedI9XUk3R2ZAe280ZQVRL88LMBP4NDt+//PiE
50n6A2g=
=4MFI

```

```

(kali㉿kali)-[~/vaje14]
$ gpg --decrypt message.txt.asc > decrypted_message.txt
gpg: encrypted with rsa4096 key, ID E8DC4A2E3661834E, created 2025-12-17
"ZanS Test (KV-Vaje) <zans@kv.test>"
gpg: Signature made Wed 17 Dec 2025 10:51:14 AM EST
gpg: using RSA key 9EEDFB1542D4D8D0916EB1A18DF495E7B56EC5CD
gpg: Good signature from "ZanS Test (KV-Vaje) <zans@kv.test>" [ultimate]

(kali㉿kali)-[~/vaje14]
$ cat decrypted_message.txt
To: peer@example.com
From: student@example.com
Date: Wed Dec 17 10:48:54 AM EST 2025
Secret message: Zaupno sporocilo

(kali㉿kali)-[~/vaje14]
$ █

```

```

277 ssh ban@112.200.111.100:2222 -p 2220
278 sudo apt update
279 sudo apt install gnupg
280 gpg --version
281 gpg --full-generate-key
282 gpg --list-keys
283 gpg --armor --export zans@kv.test > zans_pubkey.asc
284 pwd
285 ls
286 mkdir vaje14
287 mv zans_pubkey.asc vaje14
288 cd vaje14
289 ls
290 cat zans_pubkey.asc
291 history
292 gpg --full-generate-key
293 gpg --list-keys
294 echo "To: peer@example.com\nFrom: student@example.com\nDate: $(date)\nSecret message: Zaupno sporočilo" > message.txt
295 cat message.txt
296 gpg --encrypt --sign --armor --recipient zans@kv.test message.txt
297 cat message.txt.asc
298 cat message.txt.asc
299 gpg --decrypt message.txt.asc > decrypted_message.txt
300 cat decrypted_message.txt
301 gpg --fingerprint

```

(kali@kali)-[~/vaje14]
\$

Odgovori:

1. Šifriranje zagotavlja zaupnost podatkov, podpis pa zagotavlja avtentičnost in integriteto.
2. Javni ključ se uporablja za šifriranje ali preverjanje podpisa, zasebni pa za dešifriranje ali ustvarjanje podpisa. Zasebni je tisti ki mora ostati skrit.
3. Po spremembi se ob dešifriranju ali preverjanju podpisa zazna napaka, in pomeni da je datoteka spremenjena oz. poškodovana.